

On the equation $x^4y^3 = z^2 + 1$

Elementary restrictions and the obstacle to infinitude

Status of the requested conclusion. I have not established infinitely many integer solutions. The arguments below prove several unconditional restrictions, including the impossibility of a nonconstant polynomial family. They also prove that the *abc* conjecture would imply *finitely* many integer solutions. Consequently, an infinitude proof would disprove that conjecture. This is not an unconditional disproof of infinitude. Nor is a complete classification established here.

The solutions $(1, 1, 0)$ and $(-1, 1, 0)$ are immediate. All assertions below are proved from the stated assumptions; no internet search or external mathematical sources were used.

1. Intuition: rational formulas versus integer solutions

An apparently promising substitution is $u = xy$. It gives the exact rational parametrization

$$x = \frac{t^2 + 1}{u^3}, \quad y = \frac{u^4}{t^2 + 1}, \quad z = t, \quad (u, t \in \mathbb{Q}, u \neq 0).$$

Indeed, multiplying the fourth power of the first expression by the cube of the second gives $t^2 + 1$. Conversely, every rational solution is obtained by taking $u = xy$ and $t = z$.

For *integers* u, t , however, integrality of both coordinates requires

$$u^3 \mid t^2 + 1 \quad \text{and} \quad t^2 + 1 \mid u^4.$$

Thus the rational parametrization leaves the essential arithmetic difficulty intact. Equivalently, in any integer solution, each prime exponent in $z^2 + 1$ has the form $4a + 3b$ with $a, b \geq 0$. A positive exponent is therefore at least 3; exponents 1, 2, 5 are impossible. This scarcity of distinct prime factors is the reason the *abc* obstruction in Section 4 appears.

Proposition 1 (Elementary restrictions). *Every integer solution has $x \neq 0$, $y > 0$, x, y odd, and z even. Every prime dividing xy is $1 \pmod{4}$, and $y \equiv 1$ or $13 \pmod{16}$. If y is a square, the solution is $(\pm 1, 1, 0)$.*

Proof. Positivity of $z^2 + 1$ forces $x \neq 0$ and $y > 0$. An even x or an even y would give $z^2 \equiv -1 \pmod{4}$, which is impossible. Therefore x, y are odd and z is even. If an odd prime $p \mid xy$, then $z^2 \equiv -1 \pmod{p}$. Multiplication by z on the nonzero residues modulo p partitions them into orbits of length exactly 4: its square is multiplication by -1 , and its fourth power is the identity. Hence $4 \mid p - 1$. Modulo 16, $x^4 \equiv 1$ and $z^2 + 1 \equiv 1$ or 5 . Cubing permutes the odd residues and is its own inverse, so $y \equiv 1$ or 13 . Finally, if $y = v^2$ with $v > 0$, put $w = x^2v^3 > 0$. Then $(w - z)(w + z) = 1$, forcing $w = 1$, $z = 0$, and $|x| = v = 1$. $\square$

2. Another infinite class of candidates is excluded

The following elementary cube obstruction also rules out every $|x|$ that is a perfect cube. We include its proof so this restriction is self-contained.

Lemma 2. *The integer equation $t^2 + 1 = v^3$ has only the solution $(t, v) = (0, 1)$.*

Proof. Necessarily $v > 0$. If t were odd, $t^2 + 1 \equiv 2 \pmod{4}$, whereas a cube is $0, 1$, or $3 \pmod{4}$. Hence t is even and v is odd.

We use unique factorization in $\mathbb{Z}[i]$, justified as follows. Its norm is $N(a + bi) = a^2 + b^2$. Rounding both coordinates of α/β to nearest integers gives a Gaussian integer q with $N(\alpha - q\beta) \leq N(\beta)/2 < N(\beta)$. Thus the Euclidean algorithm and Bézout's identity hold. Bézout's identity makes every irreducible prime; induction on the norm gives factorization, and primality gives uniqueness.

In $(t + i)(t - i) = v^3$, the two factors are coprime: a common divisor divides $2i$, so its norm divides 4, while it also divides the odd integer $t^2 + 1$. Its norm is therefore 1. Unique factorization now shows that $t + i = \varepsilon(a + bi)^3$ for a unit ε . Cubing permutes the four units $1, -1, i, -i$, so the unit can be absorbed into the cube. Comparing imaginary parts gives

$$b(3a^2 - b^2) = 1.$$

Thus $b = \pm 1$. The case $b = 1$ requires $3a^2 = 2$, which is impossible; the case $b = -1$ gives $a = 0$. Consequently $t = 0$ and $v = 1$. $\square$

Corollary 3. *If $|x|$ is a perfect cube, then $(x, y, z) = (\pm 1, 1, 0)$.*

Proof. Write $|x| = a^3$ with $a > 0$. Then $z^2 + 1 = (a^4y)^3$. The lemma yields $z = 0$ and $a^4y = 1$, so $a = y = 1$. $\square$

3. Why a polynomial construction cannot work

Proposition 4. *If $X, Y, Z \in \mathbb{Q}[T]$ satisfy $X^4Y^3 = Z^2 + 1$ identically, all three polynomials are constant.*

Proof. X and Y are nonzero. If Z is constant, comparison of degrees proves the claim. Otherwise write $a = \deg X$, $b = \deg Y$, $c = \deg Z \geq 1$. Comparison of degrees gives $2c = 4a + 3b$. Differentiation gives

$$2ZZ' = X^3Y^2(4X'Y + 3XY').$$

No nonconstant polynomial divides both Z and XY : such a divisor would divide $Z^2 + 1$ and Z , hence 1. Therefore X^3Y^2 is coprime to Z , and the displayed identity implies $X^3Y^2 \mid Z'$. Taking degrees yields

$$3a + 2b \leq c - 1 = 2a + \frac{3}{2}b - 1, \quad \text{so} \quad a + \frac{1}{2}b \leq -1,$$

contradicting $a, b \geq 0$. $\square$

The intuition is that differentiation preserves too much of the repeated factors on the left, while reducing the available degree on the right.

Corollary 5 (Rational formulas at integer parameters). *No nonconstant triple of rational functions in T can yield integer solutions for infinitely many integer values of T .*

Proof. First suppose $f \in \mathbb{Q}(T)$ is integer-valued at infinitely many integers. Polynomial division writes $f = P + R/Q$, with $P, R, Q \in \mathbb{Q}[T]$, $Q \neq 0$, and either $R = 0$ or $\deg R < \deg Q$. Choose a positive integer D such that $DP \in \mathbb{Z}[T]$. Along those integer inputs, which are unbounded in absolute value, $Df - DP = DR/Q$ is an integer tending to zero. It therefore vanishes at infinitely many inputs, forcing $R = 0$. Thus f is a polynomial.

Apply this argument to each coordinate. The equation holds identically, because a nonzero rational function has only finitely many zeros. The preceding proposition then makes every coordinate constant. $\square$

This result concerns rational functions evaluated at *integer* parameters. It does not rule out recurrence sequences or other possible constructions of integer solutions, and it does not conflict with the two-parameter rational parametrization in Section 1.

4. Infinitude would contradict the *abc* conjecture

For a positive integer m , let $\text{rad}(m)$ be the product of its distinct prime divisors, with $\text{rad}(1) = 1$. The form of the *abc conjecture* needed here says that for every $\varepsilon > 0$ there is a constant K_ε such that pairwise coprime positive integers A, B, C with $A + B = C$ satisfy

$$C \leq K_\varepsilon \text{rad}(ABC)^{1+\varepsilon}.$$

This is an additional conjectural hypothesis, not an ingredient in any of the preceding unconditional results.

Proposition 6 (Conditional finiteness). *If the *abc* conjecture holds, $x^4y^3 = z^2 + 1$ has only finitely many integer solutions.*

Proof. The case $z = 0$ gives only $(\pm 1, 1, 0)$. For $z \neq 0$, put $C = z^2 + 1 = x^4y^3$. Every prime exponent in C is at least 3, so

$$\text{rad}(C)^3 \leq C.$$

Since $|z| < C^{1/2}$, we have

$$\text{rad}(z^2C) \leq |z| \text{rad}(C) \leq C^{1/2+1/3} = C^{5/6}.$$

Apply the conjecture to $1 + z^2 = C$, with $\varepsilon = 1/10$. These three integers are pairwise coprime, and thus

$$C \leq K_{1/10} \text{rad}(z^2C)^{11/10} \leq K_{1/10} C^{11/12}, \quad C \leq K_{1/10}^{12}.$$

This bounds z . Each fixed positive C has only finitely many factorizations $C = x^4y^3$ with $x \neq 0$, $y > 0$, proving the assertion. $\square$

The argument explains the obstruction quantitatively: $z^2 + 1$ would be large relative to the product of the distinct primes in $z^2(z^2 + 1)$. It supplies no unconditional global bound and no usable numerical bound without control of $K_{1/10}$.

5. A reproducible finite check

An exact integer computation gives the following bounded result:

$$|z| \leq 10^{10} \implies (x, y, z) \in \{(1, 1, 0), (-1, 1, 0)\}.$$

This is a computational verification within the stated bound, not a claim about larger solutions. The following complete Python code uses only integer arithmetic and the standard library; it checked 2,503,682 candidate pairs.

```
from math import isqrt

def icbrt(n):
    # The greatest integer r such that r**3 <= n.
    lo, hi = 0, 1 << ((n.bit_length() + 2) // 3)
    while lo + 1 < hi:
        mid = (lo + hi) // 2
        if mid**3 <= n:
            lo = mid
        else:
            hi = mid
    return lo

B = 10**10
C = B*B + 1
solutions = []
checked = 0
for x in range(1, isqrt(isqrt(C)) + 1, 2):
    ymax = icbrt(C // x**4)
    for y in range(1, ymax + 1, 4):
        n = x**4 * y**3 - 1
        z = isqrt(n)
        checked += 1
        if z*z == n:
            solutions.append((x, y, z))
print(checked, solutions)
# Output: 2503682 [(1, 1, 0)]
```

Why the check is exhaustive within its bound. Sign changes of x or z preserve the equation, so it suffices to take $x > 0$ and $z \geq 0$. If $z \leq B$, then $x^4y^3 \leq B^2 + 1 = C$. Since $y \geq 1$,

$$1 \leq x \leq \lfloor C^{1/4} \rfloor, \quad 1 \leq y \leq \left\lfloor \sqrt[3]{\lfloor C/x^4 \rfloor} \right\rfloor.$$

The outer loop uses $\lfloor C^{1/4} \rfloor = \text{isqrt}(\text{isqrt}(C))$. The function `icbrt` maintains $\text{lo}^3 \leq n < \text{hi}^3$, terminates when the endpoints are adjacent, and therefore returns the exact integer cube root. The parity and congruence restrictions justify testing only odd x and $y \equiv 1 \pmod{4}$. Finally, `isqrt` and the test $z^2 = n$ detect squares exactly. Thus every possible solution in the range is tested.

What remains unproved. This note establishes neither global finiteness nor global infinitude, and it does not prove that $(\pm 1, 1, 0)$ are the complete list without a bound on z . The requested unconditional proof of infinitely many integer solutions is therefore not supplied. The conditional obstruction and the unconditional restrictions above should not be mistaken for such a proof.